

Hytera Intrinsically Safe Two-Way Radio Catalog



Hytera Communications Corporation Limited
Stock Code: 002583.SZ
Address: Hytera Tower, Hi-Tech Industrial Park North, 9108# Beihuan Road,
Nanshan District, Shenzhen, P.R.C
Tel: +86-755-2697 2999 Fax: +86-755-8613 713 Post:518057
Https://www.hytera.com marketing@hytera.com



Hytera retains right to change the product design and specification. Should any printing mistake occur, Hytera doesn't bear relevant responsibility. Little difference between real product and product indicated by printing materials will occur by printing reason.
Hytera are registered trademarks of Hytera Communications Corporation Limited.
© 2026 Hytera Communications Corporation Limited. All Rights Reserved.



INNOVATE for a Safer World www.hytera.com

Hytera Intrinsically Safe Two-way Radios Introduction



Introduction to Intrinsically Safe Certification

IIB Intrinsically Safe Two-way Radios

- HP70X/HP78X UL913
- HP50X/HP56X UL913
- PT580H Plus UL913
- PNC460U

IIC Intrinsically Safe Two-way Radios

- HP79XEx Series
- HP71XEx Series
- PT890Ex

Intrinsically Safe Accessories

Intrinsically Safe Two-way Radio Comparison Table



Example of IIB Intrinsically Safe Certificate (HP780 UI913):

CCC

Ex ib II B T4 Gb, Ex ib IIIC T130°C Db

UL913

Class I, II, III, Division 1, Groups C-G, T4, -30 °C to 60 °C
Class I, Division 2, Groups A-D, T4, -30 °C to 60 °C

Example of IIC Intrinsically Safe Certificate (PT890Ex):

IECEX

Ex ib I Mb
Ex ib IIC T4 Gb
Ex ib IIIC T90°C Db
IP66/IP67/IP68, -25°C≤Ta≤+60°C

ATEX

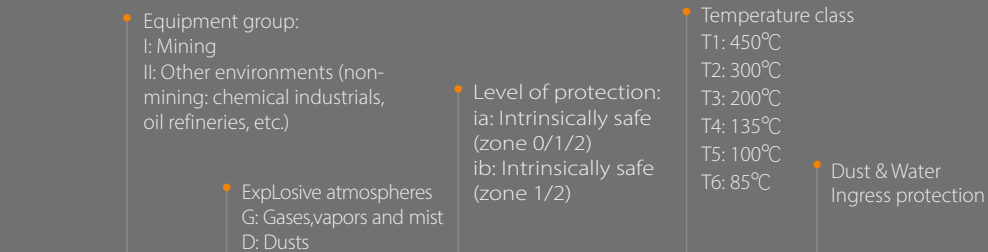
I M2 Ex ib I Mb
II 2G Ex ib IIC T4 Gb
II 2D Ex ib IIIC T90°C Db
IP66/IP67/IP68, -25°C≤Ta≤+60°C

US

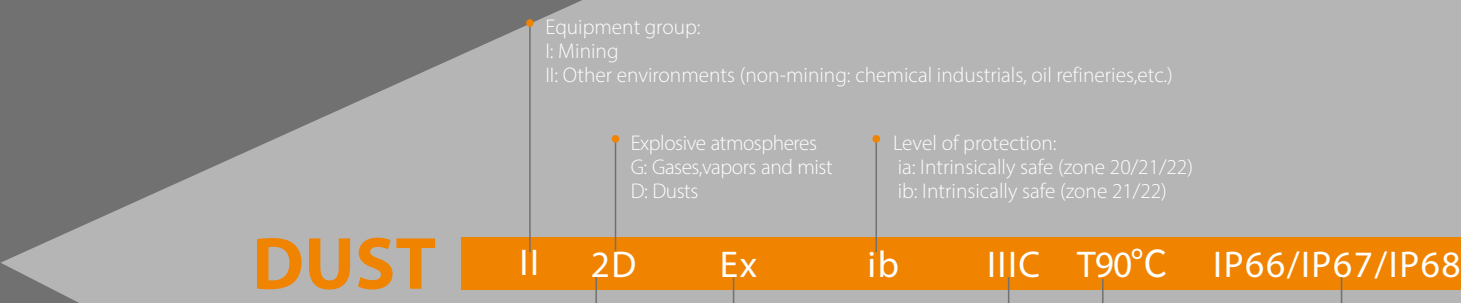
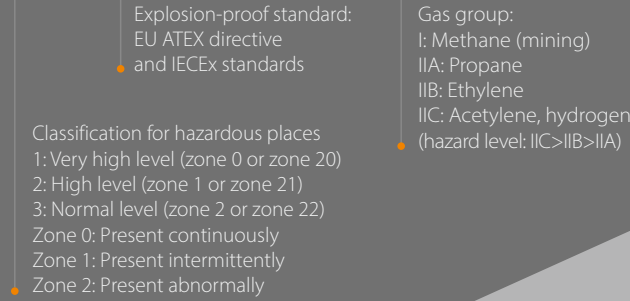
Class I, Zone 1, AEx ib IIC T4 Gb
Zone 21, AEx ib IIIC T90°C Db
IP66/IP67/IP68, -25°C≤Ta≤+60°C

CA

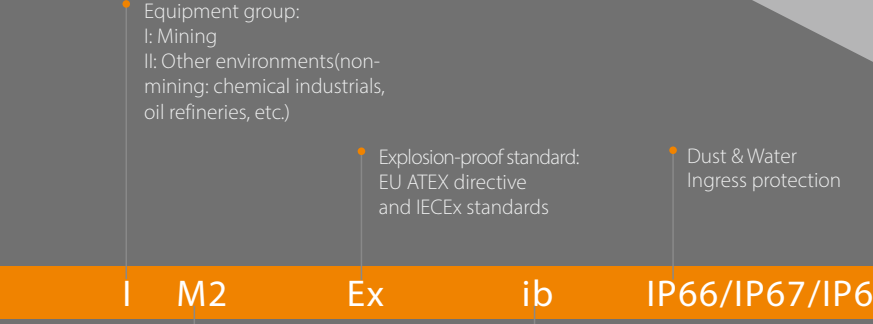
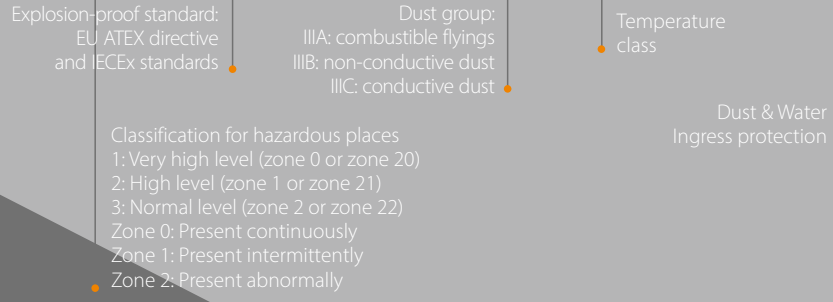
Ex ib IIC T4 Gb
Ex ib IIIC T90°C Db
IP66/IP67/IP68, -25°C≤Ta≤+60°C



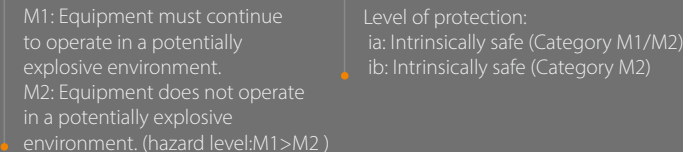
II 2G Ex ib IIC T4 IP66/IP67/IP68 **GAS**



DUST II 2D Ex ib IIIC T90°C IP66/IP67/IP68



I M2 Ex ib IP66/IP67/IP68 **MINING**





IIB Intrinsically Safe Two-way Radios

HP70X/HP78X UL913

Intrinsically Safe DMR Portable Two-way Radio

In a high-risk production environment, safe and efficient instant communication is crucial to ensuring the safety of employees and improving production efficiency. HP78X/HP70X intrinsically safe DMR portable two-way radio is an intrinsically safe explosion-proof terminal specially designed by Hytera for energy and petrochemical, mining, fire emergency, airport and other industries. Provide safe, reliable and efficient communication services in harsh flammable and explosive environments.

Features

UL913, CSA and TIA4950 Certificated Intrinsically Safe Radio

Hytera HP7 intrinsic safety radio is designed to the requirement of USA UL913, Canada CSA and TIA4950 standards. The radio works safely in hazardous environments with explosive gas and dust particles.

Smart Explosion-proof Battery

The radios support smart Explosion- Proof battery, which makes it easier to monitor the battery status, such as lifetime, charging time, etc. The Explosion-Proof Battery Identification feature helps users avoid potential safety hazards caused by the use of non-Ex batteries.

Clear and Loud Audio

The ability to clearly communicate in ambient noise-riddled places is a key requirement of all radio users. The radios have AI-based noise cancellation technology to filter unwanted background noise, delivering louder and clearer audio in background noise conditions, which was never possible before. Meanwhile, the radios equip an optimized powerful loudspeaker that can help users to hear clear sounds even in a noisy environment.

Light and Portable

The radios are in a light and compact form factor, allowing operatives to move freely and comfortably.

Extended Coverage

The HP7 UL913 series that have improved sensitivity and extended conventional coverage, allowing the setup of each call, the delivery of each message, and the report of each location in every moment that matters. The increased coverage also improves user security in a cost-effective way.

Water-porting Design

Water-porting is designed to continually shed water from the radio's audio pathways, permitting intelligible audio to be transmitted and received even when the radio is exposed to rain or water washing from the hose. Water-porting even allows clear audio immediately after being retrieved from total submersion.



HP50X/HP56X UL913

Intrinsically Safe DMR Portable Two-way Radio

When your people are working in potentially explosive environments such as chemical factory, oil refinery, firefighting and mining field, their safety is always your paramount concern. Hytera HP50X intrinsically safe digital portable radio is the right solution for your needs. From inside to outside, the HP50X/HP56X UL913 can confront any harsh conditions and offer reliable one-to-one or one-to-many communications, without compromising ease of use. Meanwhile, it delivers louder and clearer audio, extended coverage, and longer battery life. So the HP50X/HP56X UL913 empowers your people to work safely and efficiently even in the most demanding environments.

Features

UL913 Intrinsically safe

Certified by the SGS agency and in strict compliance with the UL913 and TIA4950 standards, the radio is specifically designed for use in dangerous conditions with potentially combustible dust, flammable liquids, and explosive gas. No matter how extreme the working environment is, the radio can always provide reliable and efficient communications.

Multiple safety mechanisms

The radio has a suite of proactive approaches to safety. It can intelligently detect whether the battery is anti-counterfeiting and explosion-proof, preventing the risk of fire and explosion posed by the use of non-anti-counterfeiting and non-explosion-proof battery. Also this radio provides Lone Worker and Emergency Alarm to make your people call for help quickly in case of an emergency, and the radio can send its location to the command center at the same time so that your people can be easily and accurately pinpointed thanks to the GPS and GLONASS positioning systems.

Superior audio quality

From high-power speaker to AI-based noise cancellation, as innovated and tested by Hytera Audio Lab, the radio makes your people hear and be heard clearly even in noisiest environments. The AI-based noise cancellation technology helps the radio filter out the ambient noise including breathes, extract the human voice, and eliminate the annoying howling occurred by radios that are 30 cm away from each other.

High RX sensitivity

With enhanced RF solution from Hytera RF&Antenna Lab adopted, the receiving sensitivity of every frequency band is improved up to 0.18 μV (–122 dBm). No matter whether the radio is in the area with weaker signal strength or at long range, it can always provide reliable communications for individuals or groups across your team to enhance safety and efficiency.





PT580H Plus UL913

**Intrinsically Safe TETRA
Portable Radio**

PT580H Plus UL913 Intrinsically Safe TETRA Portable Radio, compliant with IS standards of UL 913 and CSA, safely delivers clear voice calls amid noises at industrial sites where there is combustible and explosive dust or gases. As a heavy-duty radio, it is fully ruggedized to withstand shock, humidity, water or dust ingress, and other challenges. Features such as lone worker, man-down, and SOS alert make the radio an essential workplace safety equipment as well.

Features

Performance Through Innovation

The antenna in the middle position provides an Omnidirectional antenna pattern for better coverage performance. The two knobs of the portable radio are separated by the antenna, which reduces accidental activation when changing the volume channel.

Instant Communication

PT580H Plus UL913 TETRA Intrinsically Safe Portable Terminal enables quick access to TETRA networks and roaming services. Additionally, PT580H Plus UL913 provides powerful interoperability with base stations and radios of different manufacturers.

Reliable Through Design

IP68 Dust&Water Protection and Hytera ALT Test simulating over years of usage. These tests ensure PT580H Plus UL913 TETRA intrinsically safe digital portable radio will stand up to the toughest conditions and environment.

User Friendly Interface

The large TFT color display enables you to access pertinent information with ease, even under bright sunlight. As many as 20 programmable keys are configurable for quick access through one button operations.

Intrinsically Safe

Designed upon UL913 and CSA, it can work safely in most hazardous environments such as petroleum, chemical, natural gas industry. It can provide you with more secure and reliable communication support.

Longer Cycle Life

The 2400mAh Ex Li-Ion battery enables your radio to continuously work longer and the battery lifespan is longer as the charge/discharge cycles are reduced.



PNC460U

Intrinsically Safe Smart Device

The PNC460U is a UL913-certified intrinsically safe smart device with the capabilities of large-screen smartphone and push-to-talk over cellular (PoC) radio rolled into one. It provides purpose-built apps, instant PoC calls, video-based dispatch, and more to get things done in less time and with less hassle. Work safer. Work more efficient.

Features

Enhance workplace safety

The PNC460U is designed for hazardous locations with potentially explosive gas and combustible dust, in compliance with the explosion-proof requirements for use in Class I, II, III; Division 1; Groups C-G; T4 and -20°C to +60°C. So the PNC460U can always provide reliable communications and efficient services for you, even in the extreme working environments.

Android 12 OS

With the Android 12 OS, the PNC460U delivers the familiar experience and ease of use that the professionals want. More importantly, it provides an open and inclusive platform to run a wide variety of key apps, making everyday tasks easier and faster.

Precise Positioning

Thanks to the built-in dual-frequency GPS (L1+L5) chip, the PNC460U provides real-time, precise location for the dispatcher with the accuracy improved to 2 meters. Know where your devices are. Keep your personnel safe.

Professional Acoustic System

With the powerful speaker and sophisticated noise cancellation technologies, the PNC460U delivers crystal-clear audio in any noisy environment such as oil exploration sites, petrochemical plants, and refineries. This best-in-class audio promises a high-quality communication experience in any work environments.

External Antenna

Unlike the radios with the built-in antenna, the PNC460U comes with the external antenna that has a 70% higher RX capability. So even in the area with weak signal strength such as mountain and underground garage, the PNC460U still conveys accurate information without missing any words.

Powerful Camera System

The PNC460U has unmatched photography capabilities, thanks to a 50 MP rear camera HDR technology, and image stabilization algorithm. It can capture clear pictures and stable live videos in the field, and feed them to the command center in real time over the LTE or WLAN network. This is invaluable for dispatcher to know what happened, significantly improving collaboration and outcomes.





IIC Intrinsically Safe Two-way Radios

HP79XEx Series

Intrinsically Safe DMR Portable Two-way Radio

HP79XEx series Intrinsically Safe DMR Portable Two-way Radio features a secure design, durable body, and advanced ergonomics. HP79XEx series provides clear and loud voice quality, extended working hours, and long-distance communication capabilities. Compliant with stringent IECEx explosion-proof standards, it holds IECEx international certifications. This design meets global explosion-proof certification standards, providing reliable safe communication for industries such as oil and gas, fire rescue, airports, manufacturing, and more.

Features

Authoritative IS Certifications

Through the rigorous testing of the globally recognized certification organizations, HP79XEx series obtains the IECEx/ATEX certificates, to reduce the risk of accidents, and ensure the safety of users and operators.

Clear and Loud Voice

With a 2W high-power speaker, the audio is louder. Integrated with AI intelligent noise reduction algorithm, automatic gain control (AGC), and other multiple audio technologies, HP79XEx series isolates kinds of noisy backgrounds, and addresses weak sound issues, to make critical voice information clearer and more reliable.

Extended Coverage

2W TX power extends the coverage area. Critical information can be received even in remote areas, reducing coverage blind spots, enabling rapid response, and collaborating in the timely handling of critical missions.

Rocksolid & Longlasting Battery

Designed with a high-grade battery protection board and battery lock, HP79XEx series avoid potential sparks even dropped accidentally, and ensures the safe use of the battery in hazardous circumstances. Additionally, HP79XEx series prompt on screen with flashing red LED indicator, effectively addressing the security risks of non-original batteries. It also adopts a large-capacity intelligent IS battery and power-saving technology design to reduce standby power consumption, meeting the need for extended communication, and allowing seamless shift changes.

Wider range of operation temperature (in Ex area)

Thanks to new explosion-proof materials and advanced mechanical design, the HP79XEx series is built to work under temperatures from -25°C to 60°C in explosion-prone areas. With stable and even heat dissipation, the HP79XEx is engineered to work in extreme conditions, bringing extra safety and efficiency to everyday work.



HP71XEx Series

Intrinsically Safe DMR Portable Two-way Radio

HP71XEx series Intrinsically Safe DMR Portable Two-way Radio features a secure design, durable body, and advanced ergonomics. HP71XEx series provides clear and loud voice quality, extended working hours, and long-distance communication capabilities. Compliant with stringent IECEx explosion-proof standards, it holds IECEx international certifications. This design meets global explosion-proof certification standards, providing reliable safe communication for industries such as oil and gas, fire rescue, airports, manufacturing, and more.

Features

Authoritative IS Certifications

Through the rigorous testing of the globally recognized certification organizations, HP71XEx series obtains the IECEx/ATEX certificates, to reduce the risk of accidents, and ensure the safety of users and operators.

Clear and Loud Voice

With a 2W high-power speaker, the audio is louder. Integrated with AI intelligent noise reduction algorithm, automatic gain control (AGC), and other multiple audio technologies, HP71XEx series isolates kinds of noisy backgrounds, and addresses weak sound issues, to make critical voice information clearer and more reliable.

Extended Coverage

2W TX power extends the coverage area. Critical information can be received even in remote areas, reducing coverage blind spots, enabling rapid response, and collaborating in the timely handling of critical missions.

Rocksolid & Longlasting Battery

Designed with a high-grade battery protection board and battery lock, HP71XEx series avoid potential sparks even dropped accidentally, and ensures the safe use of the battery in hazardous circumstances. Additionally, HP71XEx series prompt on screen with flashing red LED indicator, effectively addressing the security risks of non-original batteries. It also adopts a large-capacity intelligent IS battery and power-saving technology design to reduce standby power consumption, meeting the need for extended communication, and allowing seamless shift changes.

Wider range of operation temperature (in Ex area)

Thanks to new explosion-proof materials and advanced mechanical design, the HP71XEx series is built to work under temperatures from -25°C to 60°C in explosion-prone areas. With stable and even heat dissipation, the HP71XEx is engineered to work in extreme conditions, bringing extra safety and efficiency to everyday work.





PT890Ex

Intrinsically Safe TETRA
Portable Radio

PT890Ex Intrinsically Safe TETRA Portable Radio features a secure design, durable body, and advanced ergonomics. PT890Ex provides clear and loud voice quality, extended working hours, and long-distance communication capabilities. Compliant with stringent IECEx explosion-proof standards, it holds IECEx international certifications. This design meets global explosion-proof certification standards, providing reliable safe communication for industries such as oil and gas, fire rescue, airports, manufacturing, and more.

Features

Authoritative IS Certifications

Through the rigorous testing of the globally recognized certification organizations, PT890Ex obtains the IECEx/ATEX certificates, to reduce the risk of accidents, and ensure the safety of users and operators.

Clear and Loud Voice

With a 2W high-power speaker, the audio is louder. Integrated with AI intelligent noise reduction algorithm, automatic gain control (AGC), and other multiple audio technologies, PT890Ex isolates kinds of noisy backgrounds, and addresses weak sound issues, to make critical voice information clearer and more reliable.

Extended Coverage

Class 3L transmission power extends the coverage area. Even in remote areas, critical information can be received, reducing coverage blind spots, enabling rapid response, and collaborating timely handling of critical tasks.

Rocksolid & Longlasting Battery

Designed with a high-grade battery protection board and a novel battery lock , PT890Ex avoid potential sparks even dropped accidentally, and ensures the safe use of the battery in hazardous circumstances. Additionally, PT890Ex prompt on screen with flashing red LED indicator, effectively addressing the security risks of non-original batteries. It also adopts a large-capacity intelligent IS battery and power-saving technology design to reduce standby power consumption, meeting the need for extended communication, and allowing seamless shift changes.

Wider range of operation temperature (in Ex area)

Thanks to new explosion-proof materials and advanced mechanical design, the PT890Ex is built to work under temperatures from –25°C to 60°C in explosion-prone areas. With stable and even heat dissipation, the PT890Ex is engineered to work in extreme conditions, bringing extra safety and efficiency to everyday work.



Intrinsically Safe Accessories

Accessories Model	Accessories Description	Compatibility Portable Radios
 ACN-01	PTT&MIC cable	PT580H Plus UL913
 ACN-02P	PTT&MIC cable	HP700/780 UL913
 EA-01	Receive-only Earpiece with Transparent Acoustic Tube	HP700/780 UL913 PT580H Plus UL913
 EH-01	Receive-Only C-Style Earloop	HP700/780 UI913 PT580H Plus UL913
 EH-02	Receive-Only Adjustable Earhook with Swivel Speaker	PT580H Plus UL913
 ES-01	Receive-Only Earbud	HP700/780 UI913 PT580H Plus UL913
 ES-02	Receive-Only Earpiece with Transparent Acoustic Tube	HP700/780 UL913
 SM26N13P-Ex	Intrinsically Safe Remote Speaker Microphone	HP79XEx PT890Ex HP71XEx
 SM18N5	Waterproof Remote Speaker Microphone(IP67)	PT580H Plus UL913
 SM26N3	Waterproof Remote Speaker Microphone(IP67)	PT580H Plus UL913
 SM26N1-P	Waterproof Remote Speaker Microphone (IP67)	HP70X/78X UL913
 SM26N2-P	Waterproof Remote Speaker Microphone(IP54)	HP70X/78X UL913
 SM24N2P-Ex	Intrinsically Safe Multifunctional Remote Speaker Microphone	HP79XEx HP71XEx PT890Ex

Accessories Model	Accessories Description	Compatibility Portable Radios
 EAN17	3-wire Surveillance Earpiece with Transparent Acoustic Tube(Beige)	PT580H Plus UL913
 EAN18	3-wire Surveillance Earpiece with Transparent Acoustic Tube(Black)	PT580H Plus UL913
 EAN23	2-wire Surveillance Earpiece with Transparent Acoustic Tube	PT580H Plus UL913
 ECN18	Heavy duty Noise-cancelling Headset	PT580H Plus UL913
 EHN16	C-Earset	PT580H Plus UL913
 EHN17	Swivel Earset	PT580H Plus UL913
 ESN12	ESN12 Earbud	PT580H PLUS UL913
 EAN21-P	3-wire Surveillance Earpiece with Transparent Acoustic Tube(Beige)	HP70X/78X UL913
 ECN21-P	Heavy duty Noise-cancelling Headset	HP70X/78X UL913
 EHN26-P	C-Earset	HP70X/78X UL913
 MCL39	Multi-unit Charger for Li-ion Battery(2A)	HP70X/78X UL913
 EHN38P-Ex	Intrinsically Safe and Adjustable Earset	HP79XEx HP71XEx PT890Ex
 POA63P-Ex	Large PTT	HP79XEx HP71XEx PT890Ex

IIB Intrinsically Safe Two-way Radio Comparison Table

	HP70X UL913		HP78X UL913	HP50X UL913	HP56X UL913	PT580H PLUS UL913	PNC460U
Frequency Range	VHF: 136~174MHz UHF: 350~470MHz		400-470MHz 136-174MHz		380-430 MHz; 405~475 MHz; 806-870 MHz		GSM: 850/900/1800/1900 WCDMA: B1/B2/B3/B4/B5/B8 TDD-LTE: B34/B38/B39/ B40/B41(2496~2690 MHz) FDD-LTE: B1/B2/B3/B4/B5/B7/ B8/B12/B13/B17/B20/B26/B28
Battery	2850mAh		2,400 mAh Li-ion		2400 mAh Li-Ion battery		4,580 mAh
Battery Life(5/5/90 Duty Cycle,High TX Power)	28h(GNSS off) 24h(GNSS on)		Digital: 23h (GPS off) Digital: 21h (GPS on)		>24 hours		53 hours
Frequency Stability	±0.5ppm		±0.5ppm		/		/
Dimensions (H x W x D)	132mm x 55mm x 36.5mm		119 mm x 55 mm x 33.5 mm		127.5 mm x 55 mm x 43 mm		164 mm × 77 mm × 23 mm
Weight(with antenna&battery)	369g	389g		300g (±5g)	321g	408 g (with 2400 mAh standard battery and antenna)	About 337g
Display	OLED Display: 0.91 inch	LCD Display: 240x320 pixel, 262000 colours, 2.4-inch, 10 line		/	1.45 inch LCD, 240x320 pixels, colorful	1.8 inch,160x128 pixels, 262144colors	5.0 inches
Anti-explosion levels	Class I, II, III, Division 1, Groups C-G, T4, -30°C to 60°C Class I, Division 2, Groups A-D, T4, -30°C to 60°C		Class I, Division 1, Group C-D, T4; Class II, Class III, Division 1, Group E-G, T135°C; Class I, Division 2, Groups A - D, T4; Ta: -20°C to 60°C		Class I, II, III, Division 1, Groups C-G, T4, -30°C to 55°C Class I, Division 2. Groups A-D T4 -30°C to 55°C		Class I, II, III; Division 1; Groups C–G; T4 and -20°C to +60°C
BT	Optional		BT 5.3		Optional		BT 5.0
Positioning	Optional		Support		Optional		Support
NFC	/		/		/		Available
WLAN	/		/		/		2.4 GHz&5 GHz, 802.11 a/b/g/n/ac
Camera	/		/		/		Rear: 50 MP; Front: 8 MP
RF Power Output	UHF: 4W/1W VHF: 5W/1W		UHF: 4W/1W VHF: 5W/1W		1W		/
Sensitivity(Digital)	0.18μV/BER5%		0.18μV/BER5%		≤ -116 dBm(typical-118 dBm) 806-870 MHz: ≤-115 dBm (typical-116 dBm)		/
Sensitivity(Analog)	0.18μV (12dB SINAD); 0.16μV (Typical) (12dB SINAD)		0.18μV (12dB SINAD); 0.16μV (Typical) (12dB SINAD)		/		/
Dust and Water Intrusion	IP68(2m4h)		IEC60529-IP67		IP65/66/67/68(2m,4h) Per IEC60529		IP68
Standard Compliance	Analog Conventional Mode Digital Conventional Mode DMR Tier III Trunking		Analog Conventional Mode Digital Conventional Mode		/		4G/3G/2G
E2E Basic Encryption	Support		Support		Support		End-to-End encryption
E2E Advanced Encryption	Optional		/		Support		/
Hard Encryption	Optional		/		/		/
Man-down	Support		/		Support		Support
Lone Worker	Support		Support		Support		Support
Voice Recording	Optional		/		/		Support
Dedicated Emergency Button	Support		/		Support		Support
Priority Interrupt	Support		Optional		Support		Support
Roaming / Site Search	Optional		Optional		Support		Support

IIC Intrinsically Safe Two-way Radio Comparison Table

	HP79XEX IIC	HP71XEX IIC	PT890EX
Frequency Range	UHF: 400-480MHz 350-400MHz 330-400MHz VHF: 136-174MHz	UHF: 400-480MHz 350-400MHz 330-400MHz VHF: 136-174MHz	380-475 MHz 806-870 MHz
Battery	2150mAh	2150mAh	2150mAh
Battery Life(5/5/90 Duty Cycle,High TX Power)	24h (GNSS OFF); 21h (GNSS ON)	24h (GNSS OFF); 21h (GNSS ON)	TMO: 26H DMO: 20H
Frequency Stability	±0.5ppm	±0.5ppm	/
Dimensions (H x W x D)	130mm x 55mm x 37mm	130mm x 55mm x 37mm	130mm x 55mm x 37mm
Weight(with antenna&battery)	about 390g	about 390g	about 390g
Display	LCD,320x240 pixel, 262000 colors, 2.4inch	LCD,172x320 pixel, 262000 colors, 1.47inch	LCD,320x240 pixel, 262000 colors, 2.4inch
Anti-explosion levels	IECEx: Ex ib I Mb Ex ib IIC T4 Gb Ex ib IIIC T120°C Db Tamb = -25°C to +60°C IP66; IP67; IP68 ATEX: I M2 Ex ib I Mb II 2G Ex ib IIC T4 Gb II 2D Ex ib IIIC T120°C Db Tamb = -25°C to +60°C IP66; IP67; IP68 USA: CL I, ZN 1, AEx ib IIC T4 Gb ZN 21, AEx ib IIIC T120°C Db Tamb = -25°C to +60°C IP66; IP67; IP68 CA: Ex ib IIC T4 Gb Ex ib IIIC T158°C Db Tamb = -25°C to +60°C IP66; IP67; IP68	IECEx: Ex ib I Mb Ex ib IIC T4 Gb Ex ib IIIC T120°C Db Tamb = -25°C to +60°C IP66; IP67; IP68 ATEX: I M2 Ex ib I Mb II 2G Ex ib IIC T4 Gb II 2D Ex ib IIIC T120°C Db Tamb = -25°C to +60°C IP66; IP67; IP68 USA: CL I, ZN 1, AEx ib IIC T4 Gb ZN 21, AEx ib IIIC T120°C Db Tamb = -25°C to +60°C IP66; IP67; IP68 CA: Ex ib IIC T4 Gb Ex ib IIIC T158°C Db Tamb = -25°C to +60°C IP66; IP67; IP68	IECEx: Ex ib I Mb Ex ib IIC T4 Gb Ex ib IIIC T90°C Db Tamb = -25°C to +60°C IP66; IP67; IP68 ATEX: I M2 Ex ib I Mb II 2G Ex ib IIC T4 Gb II 2D Ex ib IIIC T120°C Db Tamb = -25°C to +60°C IP66; IP67; IP68 USA: CL I, ZN 1, AEx ib IIC T4 Gb ZN 21, AEx ib IIIC T90°C Db Tamb = -25°C to +60°C IP66; IP67; IP68 CA: Ex ib IIC T4 Gb Ex ib IIIC T190°C Db Tamb = -25°C to +60°C IP66; IP67; IP68
BT	BT 5.3	BT 5.3	BT 5.3
Positioning	Support	Support	Support
NFC	Available	Available	Available
WLAN	2.4 G	2.4 G	2.4 G
Camera	/	/	/
RF Power Output	2W/1W	2W/1W	Class (max)
Sensitivity(Digital)	0.16uV/BER5%	0.16uV/BER5%	-120dBm(380-475MHz) -119dBm(806-870MHz)
Sensitivity(Analog)	0.16uV(12dB SINAD) 0.14uV(Typical)(12dB SINAD)	0.16uV(12dB SINAD) 0.14uV(Typical)(12dB SINAD)	/
Dust and Water Intrusion	IP64/IP65/IP66/IP67/IP68 PER IEC-60079-0:2017 & IEC-60529	IP64/IP65/IP66/IP67/IP68 PER IEC-60079-0:2017 & IEC-60529	IP64/IP65/IP66/IP67/IP68 PER IEC-60079-0:2017 & IEC-60529
Standard Compliance	Analog Conventional Mode Digital Conventional Mode MPT DMR Tier III Trunking	Analog Conventional Mode Digital Conventional Mode MPT DMR Tier III Trunking	/
E2E Basic Encryption	Support	Support	Support
E2E Advanced Encryption	Optional	Optional	Optional
Hard Encryption	Optional	Optional	/
Man-down	Support	Support	Support
Lone Worker	Support	Support	Support
Voice Recording	Optional	Optional	Support
Dedicated Emergency Button	Support	Support	Support
Priority Interrupt	Support	Support	Support
Roaming / Site Search	Optional	Optional	Optional

IIA Intrinsically Safe Two-way Radio Comparison Table

	HP79XEX IIA	HP71XEX IIA
Frequency Range	UHF: 400-480MHz 350-400MHz 330-400MHz VHF: 136-174MHz	UHF: 400-480MHz 350-400MHz 330-400MHz VHF: 136-174MHz
Battery	2150mAh	2150mAh
Battery Life(5/5/90 Duty Cycle,High TX Power)	22h (GNSS OFF); 19h (GNSS ON)	22h (GNSS OFF); 19h (GNSS ON)
Frequency Stability	±0.5ppm	±0.5ppm
Dimensions (H x W x D)	130mm x 55mm x 37mm	130mm x 55mm x 37mm
Weight(with antenna&battery)	about 390g	about 390g
Display	LCD,320x240 pixel, 262000 colors, 2.4inch	LCD,172x320 pixel, 262000 colors, 1.47inch
Anti-explosion levels	IECEx: Ex ib I Mb Ex ib IIA T4 Gb Ex ib IIIC T120°C Db Tamb = -25°C to +50°C IP66; IP67; IP68 ATEX: I M2 Ex ib I Mb II 2G Ex ib IIA T4 Gb II 2D Ex ib IIIC T120°C Db Tamb = -25°C to +50°C IP66; IP67; IP68 USA: CL I, ZN 1, AEx ib IIA T4 Gb ZN 21, AEx ib IIIC T120°C Db Tamb = -25°C to +50°C IP66; IP67; IP68 CA: Ex ib IIA T4 Gb Ex ib IIIC T158°C Db Tamb = -25°C to +50°C IP66; IP67; IP68	IECEx: Ex ib I Mb Ex ib IIA T4 Gb Ex ib IIIC T120°C Db Tamb = -25°C to +50°C IP66; IP67; IP68 ATEX: I M2 Ex ib I Mb II 2G Ex ib IIA T4 Gb II 2D Ex ib IIIC T120°C Db Tamb = -25°C to +50°C IP66; IP67; IP68 USA: CL I, ZN 1, AEx ib IIA T4 Gb ZN 21, AEx ib IIIC T120°C Db Tamb = -25°C to +50°C IP66; IP67; IP68 CA: Ex ib IIA T4 Gb Ex ib IIIC T158°C Db Tamb = -25°C to +50°C IP66; IP67; IP68
BT	BT 5.3	BT 5.3
Positioning	Support	Support
NFC	Available	Available
WLAN	2.4 G	2.4 G
Camera	/	/
RF Power Output	UHF: 4W/1W VHF: 5W/1W	UHF: 4W/1W VHF: 5W/1W
Sensitivity(Digital)	0.16uV/BER5%	0.16uV/BER5%
Sensitivity(Analog)	0.16uV(12dB SINAD) 0.14uV(Typical)(12dB SINAD)	0.16uV(12dB SINAD) 0.14uV(Typical)(12dB SINAD)
Dust and Water Intrusion	IP64/IP65/IP66/IP67/IP68 PER IEC-60079-0:2017 & IEC-60529	IP64/IP65/IP66/IP67/IP68 PER IEC-60079-0:2017 & IEC-60529
Standard Compliance	Analog Conventional Mode Digital Conventional Mode MPT DMR Tier III Trunking	Analog Conventional Mode Digital Conventional Mode MPT DMR Tier III Trunking
E2E Basic Encryption	Support	Support
E2E Advanced Encryption	Optional	Optional
Hard Encryption	Optional	Optional
Man-down	Support	Support
Lone Worker	Support	Support
Voice Recording	Optional	Optional
Dedicated Emergency Button	Support	Support
Priority Interrupt	Support	Support
Roaming / Site Search	Optional	Optional

